

BUBR1-RESCUE 3.0

Genotype-Guided Drug Repurposing Framework for BUB1B/BUBR1-Associated Mosaic Variegated Aneuploidy

Rare Disease Real Kid Hackathon 2026 - Track 2

Track 2 Final Report | Computational and literature-guided analysis

Scope statement: This project does not claim a validated treatment. It prioritizes falsifiable rescue hypotheses and separates direct evidence from mechanistic inference. No original wet-lab rescue experiment was completed.

Abstract

BUBR1-RESCUE 3.0 is a genotype-guided drug-repurposing framework for BUB1B-associated Mosaic Variegated Aneuploidy (MVA). The analysis combines variant interpretation, structure-informed rescue reasoning, a reproducible 20-candidate screen, evidence-strength grading, and explicit Go/No-Go validation logic. One allele, BUB1B p.Leu737Ter, is treated as a pathogenic loss-of-function allele. A second missense variant, p.Asn1002Lys (N1002K), remains of unresolved pathogenicity and unresolved phase. The project therefore does not claim a confirmed compound-heterozygous diagnosis. Instead, it asks whether residual full-length BUBR1, if present, could be molecularly rescued. Niacin emerged as the primary repurposing hypothesis through the NAD⁺-SIRT2-BUBR1 axis, NMN serves as a mechanistic benchmark, and arimoclomol provides an orthogonal proteostasis hypothesis. The framework is designed to be falsifiable: a candidate advances only if it improves both BUBR1 abundance/stability and chromosome-segregation function. No wet-lab rescue was demonstrated.

1. Background and scientific question

BUBR1, encoded by BUB1B, is a key component of the spindle assembly checkpoint and chromosome-segregation machinery. BUB1B-associated MVA can arise through combinations of truncating and missense alleles that reduce functional BUBR1 abundance or impair BUBR1 activity. Prior patient-cell work demonstrated that some MVA-associated BUBR1 missense variants are unstable and that restoring protein abundance can restore checkpoint function in selected contexts [1].

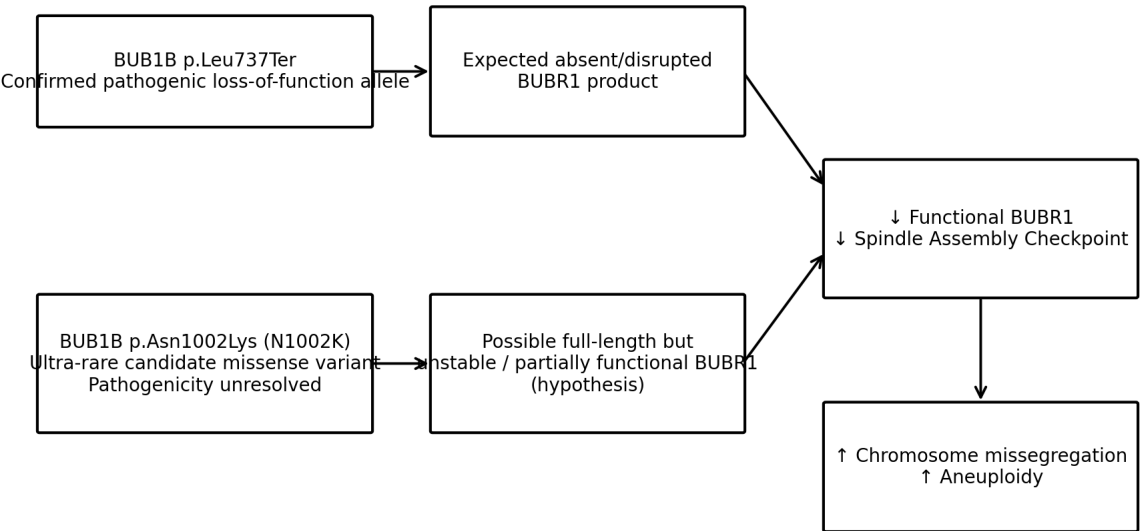
Central question: Can a case with one established loss-of-function allele and one unresolved missense candidate be converted into a transparent, testable drug-repurposing strategy without overstating genetic certainty?

2. Genomic interpretation

Variant	Interpretation	Protein consequence	Project confidence	Critical caveat
p.Leu737Ter	Pathogenic loss-of-function allele	Premature stop at residue 737	High	Variant-level pathogenicity does not establish phase with N1002K
p.Asn1002Lys (N1002K)	Candidate missense variant	Full-length protein remains possible	Unresolved causality; moderate-high priority for rescue testing	Pathogenicity and phase remain unresolved

ClinVar currently lists NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter) as pathogenic for MVA syndrome 1 [2]. N1002K is not treated here as proven pathogenic. The project deliberately separates causality confidence from rescuability plausibility.

BUBR1-RESCUE 3.0 — Variant → Molecular Defect → Phenotype



Constraint: phase and N1002K pathogenicity are unresolved; no compound-heterozygous diagnosis is claimed.

Figure 1. Variant-to-phenotype model used in BUBR1-RESCUE 3.0. N1002K remains a hypothesis-generating variant, not a confirmed disease-causing allele.

3. Structure-informed rescuability reasoning

N1002K lies in the C-terminal kinase-like/pseudokinase region of BUBR1. The substitution changes a neutral polar asparagine to a positively charged lysine, making changes in local packing,

electrostatics, or hydrogen-bonding mechanistically plausible. However, structural plausibility is not equivalent to pathogenicity or experimental destabilization.

- Stability-defect candidate: moderate support.
- Intrinsic functional-defect candidate: unresolved.
- Benign or minimal structural effect: cannot be excluded.
- No N1002K-specific abundance, folding, checkpoint, or rescue assay was completed.

The strongest disease-mechanism precedent comes from prior MVA work showing that missense variants such as I909T and L1012P can show reduced BUBR1 abundance, increased turnover, and dependence on HSP90-assisted folding. When selected unstable mutants were expressed to wild-type-like abundance, checkpoint function could be restored [1]. This supports testing a rescue hypothesis, but it does not establish that N1002K shares the same mechanism.

4. Variant-Rescue framework

The key conceptual change in BUBR1-RESCUE 3.0 is the explicit separation of two questions:

- Variant causality: How strong is the evidence that the variant contributes to disease?
- Molecular rescuability: If the variant is biologically relevant, is residual protein abundance or function potentially rescuable?

For N1002K, causality remains unresolved, while rescuability is considered mechanistically plausible enough to justify falsifiable testing. This prevents a common error in rare-disease hypothesis generation: treating a computationally interesting variant as if it were already proven pathogenic.

5. Systematic drug-repurposing screen

Twenty candidate compounds or pathway strategies were evaluated using an internal 100-point prioritization framework. The score is not clinically validated and does not predict efficacy. It is a transparent project tool designed to make selection criteria explicit and reproducible.

Rank	Candidate	Project role	Score / 100
1	Niacin	Primary repurposing hypothesis	84
2	NMN	Mechanistic benchmark	83
3	Arimoclomol	Orthogonal proteostasis hypothesis	79
4	Nicotinamide riboside	Backup NAD ⁺ strategy	75
5	GGA / teprenone	Backup HSP strategy	64
6	Bimoclomol	Backup proteostasis strategy	63
7	TUDCA	Exploratory	59

		comparator	
8	4-PBA	Comparator	55

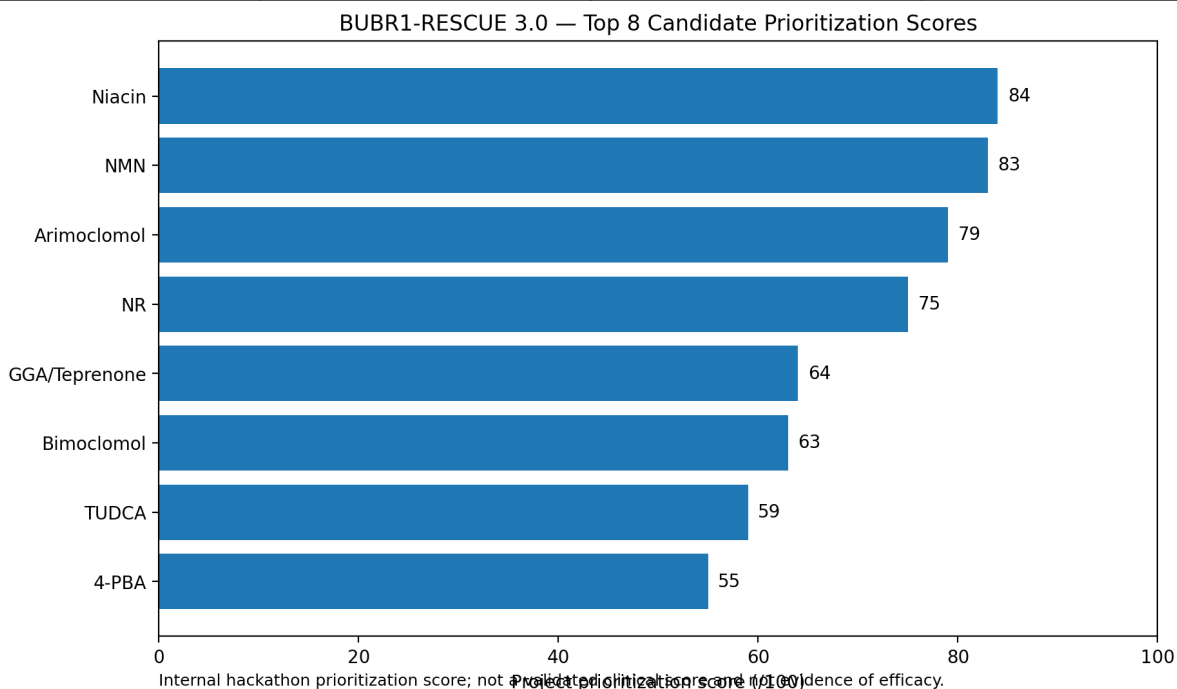


Figure 2. Top eight project-prioritization scores. These scores rank hypotheses for testing; they are not clinical efficacy scores.

6. Evidence-strength analysis

6.1 Primary hypothesis: niacin / NAD⁺-SIRT2

Nicotinic acid can increase cellular NAD⁺ through NAPRT-dependent metabolism in human cells [3]. Separately, SIRT2 is an NAD⁺-dependent deacetylase, and experimental work has linked SIRT2 to BUBR1 abundance. North et al. reported that SIRT2 maintains BubR1 in a deacetylated state and that NMN treatment increased BubR1 abundance in vivo [4].

What is direct: niacin → NAD⁺; NAD⁺/SIRT2 → BUBR1 abundance; NMN → increased BubR1 in vivo.

What remains inferential: niacin → rescue of BUBR1 N1002K or MVA. No direct evidence for that step was identified.

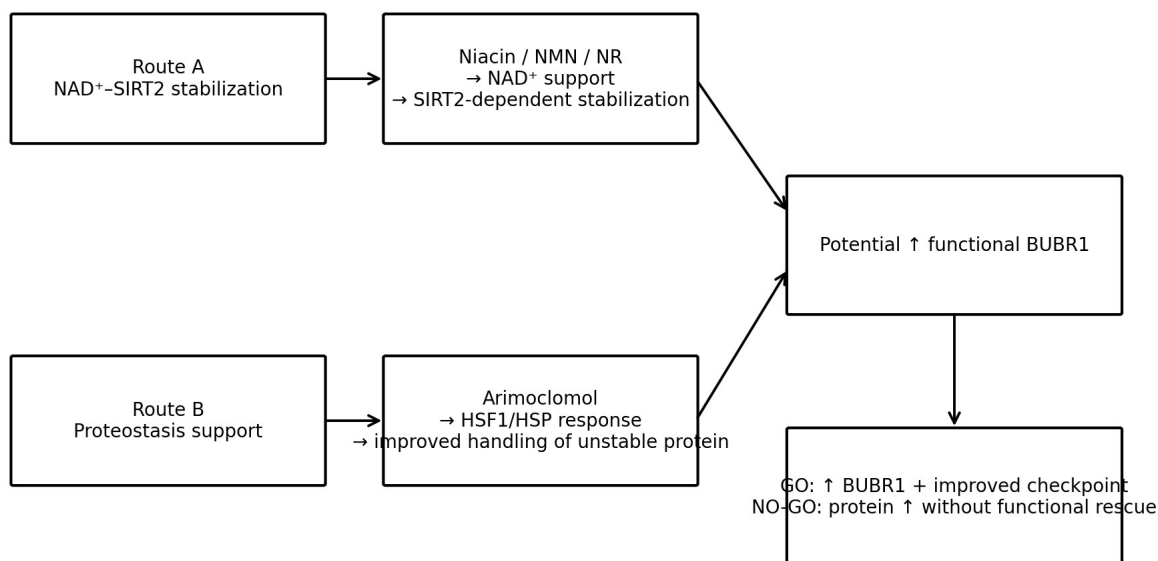
6.2 Orthogonal hypothesis: arimoclomol / proteostasis

Arimoclomol provides an independent proteostasis route. The logic is supported indirectly by disease-relevant evidence: unstable MVA-associated BUBR1 missense proteins can depend on heat-shock machinery, while arimoclomol has been developed as a heat-shock-response co-inducer. Arimoclomol is FDA-approved as MIPLYFFA, in combination with miglustat, for neurological manifestations of Niemann-Pick disease type C in patients 2 years of age and older [5]. The FDA

label also notes that the mechanism by which arimoclomol exerts clinical effects in NPC is unknown [5].

Critical gap: there is no direct evidence that arimoclomol stabilizes BUBR1 or rescues N1002K.

BUBR1-RESCUE 3.0 — Dual-Pathway Rescue Strategy



These are testable rescue hypotheses, not demonstrated treatments for N1002K or MVA.

Figure 3. Dual-pathway rescue strategy. Route A targets NAD⁺-SIRT2-linked BUBR1 stabilization; Route B tests a proteostasis hypothesis.

7. Negative evidence and candidate rejection

A useful prioritization framework must be able to reject candidates, not only nominate them. BUBR1-RESCUE 3.0 therefore retains mechanistically unfavorable or low-priority compounds as explicit controls.

- Geldanamycin: contra-mechanistic. HSP90 inhibition reduced levels of unstable BUBR1 missense proteins in prior MVA work [1].
- MG132: research-only proteasome tool, not a repurposing candidate.
- Nicotinamide: mechanistically conflicted for a SIRT2-dependent rescue strategy because nicotinamide can inhibit sirtuins.
- 4-PBA: retained as a comparator rather than a lead because target specificity is weak and its conventional chemical-chaperone interpretation is not sufficient for a BUBR1-specific claim.

8. Final decision framework and falsifiability

BUBR1-RESCUE 3.0 — Final Decision Framework

Core principle: prioritize testable rescue strategies without overstating variant pathogenicity or treatment efficacy.

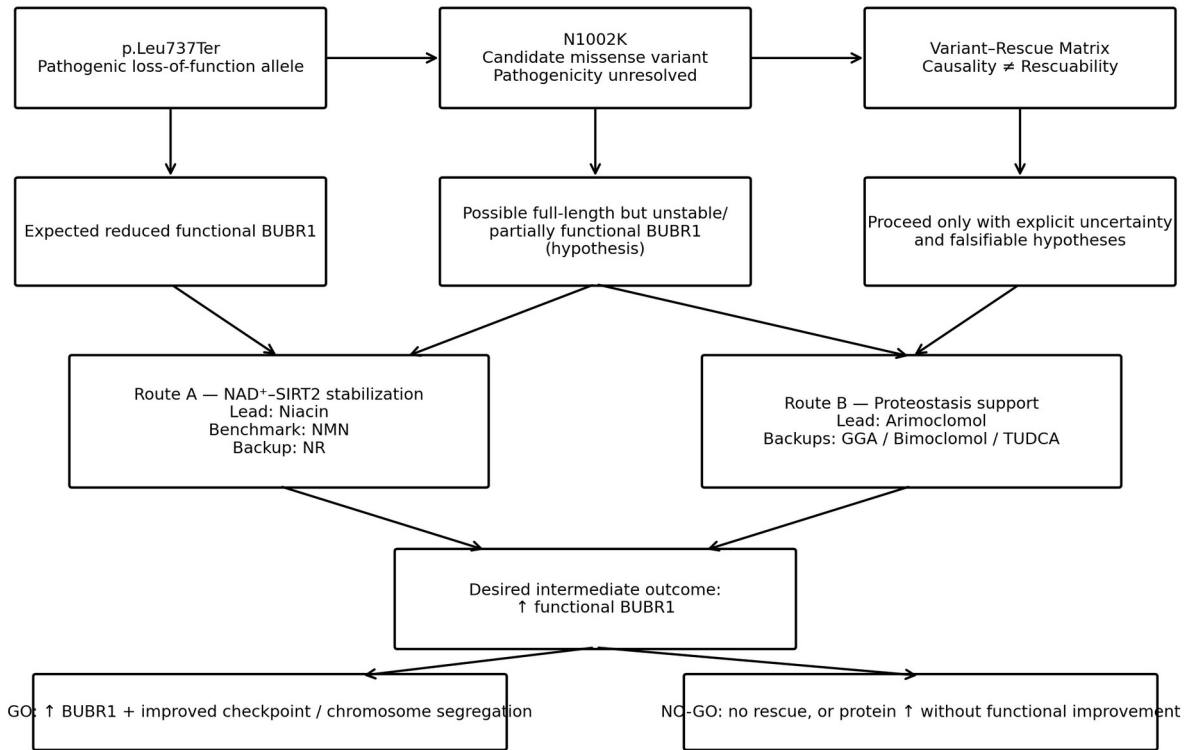


Figure 4. Final decision framework. A candidate advances only if molecular rescue and functional rescue occur together.

GO criterion: increased BUBR1 abundance or stability together with improved checkpoint/chromosome-segregation function.

NO-GO criterion: no molecular rescue, protein increase without functional rescue, or worsening chromosome-segregation phenotypes.

9. Stress test: what would falsify the hypotheses?

- Niacin route fails if NAD⁺ increases but BUBR1 abundance does not.
- Niacin route also fails if BUBR1 abundance increases but checkpoint function or chromosome segregation does not improve.
- Arimoclomol route loses its mechanistic basis if N1002K does not create a folding/stability or proteostasis-sensitive defect.
- Any route is rejected if molecular improvement occurs without functional rescue.

10. Limitations

- N1002K pathogenicity is unresolved.

- Phase between p.Leu737Ter and N1002K is unresolved.
- No patient-specific BUBR1 abundance or functional checkpoint assay was performed.
- No N1002K-specific stability, folding, or rescue assay was performed.
- Niacin has not been shown to rescue BUBR1 N1002K.
- Arimoclomol has not been shown to rescue BUBR1 N1002K.
- The internal drug-ranking score is not a clinical score and should not be interpreted as treatment efficacy.
- No dose, regimen, or patient treatment recommendation is made.

11. Conclusion

BUBR1-RESCUE 3.0 prioritizes niacin as a testable NAD⁺-SIRT2 repurposing hypothesis, NMN as a mechanistic benchmark, and arimoclomol as an orthogonal proteostasis hypothesis. The framework does not claim therapeutic efficacy. Its contribution is a transparent, evidence-graded, reproducible and falsifiable path for asking whether residual BUBR1 can be molecularly rescued while preserving uncertainty about N1002K pathogenicity and phase.

12. AI-use disclosure

OpenAI ChatGPT, Plus plan, was used for literature organization, hypothesis structuring, drafting, code/document preparation, and interpretation support. The account-level "Improve the model for everyone" setting was enabled during the analysis. Raw FASTQ files and the complete genome-wide VCF were not uploaded to ChatGPT; genomic analysis was performed locally, and only targeted, non-reconstructive variant summaries and selected annotation outputs were provided for assistance.

AI-generated outputs were treated as supportive material and were independently checked against primary literature, public variant databases, and the challenge data before inclusion in the final interpretation.

13. Reproducibility and repository outputs

- data/drug_candidate_ranking.csv - candidate-level scoring matrix
- data/evidence_strength_matrix.csv - claim-by-claim evidence classification
- docs/drug_ranking_method.md - ranking methodology
- docs/evidence_strength_analysis.md - direct vs inferential evidence analysis
- docs/claim_audit.md - safe, qualified and prohibited claims
- docs/limitations.md - explicit project limitations
- results/summary_results.md - consolidated results
- results/final_decision_framework.md - Go/No-Go decision logic
- figures/ - visual summaries used in this report

References

- [1] Suijkerbuijk SJE, et al. Molecular causes for BUBR1 dysfunction in the human cancer predisposition syndrome Mosaic Variegated Aneuploidy. *Cancer Research*. 2010. PMCID: PMC2887387.
- [2] NCBI ClinVar. NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter) and Mosaic variegated aneuploidy syndrome 1. RCV000641226. Current record; last evaluated Oct 9, 2024.
- [3] Hara N, et al. Elevation of cellular NAD levels by nicotinic acid and involvement of nicotinic acid phosphoribosyltransferase in human cells. *Journal of Biological Chemistry*. 2007. PMID: 17604275.
- [4] North BJ, et al. SIRT2 induces the checkpoint kinase BubR1 to increase lifespan. *EMBO Journal*. 2014;33(13):1438-1453. PMID: 24825348. PMCID: PMC4194088.
- [5] U.S. Food and Drug Administration. MIPLYFFA (arimoclomol) prescribing information. Initial U.S. approval 2024. NDA 214927.